Guided Assignment 4

Decisions, memory of the past, learning, and what a population can compute

PN4 · Lectures 7, 8, 9 (and the Lecture 6 integrator)

Purpose. Build, by hand, the reasoning that PN4 says Test 4 assumes: "everything above by hand or from scratch." The emphasis is on mechanisms, sketches and short derivations, not on long arithmetic. This assignment is for learning. It does not predict the real test.

How to work. For every part: *predict* (write a one-line guess or rough sketch), then *derive or draw*, then *explain* in one or two sentences. Use pencil and paper, with no calculator and no code. Check the hints page of the companion only after a genuine attempt, and check a stage's worked solutions only after finishing that stage.

Pacing for a one-evening push (about 2.5–3 h). The core stages are **1, 3, 4, 5, 7**, each about 25–30 minutes. Stages **2** and **6** are shorter, about 10–15 minutes each. If you run out of time, skim them using the solutions. Save Stage 8 and the practice test for a fresh, timed sitting.

Conventions. Time is in seconds. Matched first-order synapses $H(s) = 1/(1+\tau s)$ implement $\dot{\mathbf{x}} = A\mathbf{x}+B\mathbf{u}$ with recurrent transform $A' = \tau A + I$ and input transform $B' = \tau B$. The error for PES is $\varepsilon = \hat{y} - y^d$ (output minus desired), which matches the notebook and Lecture 8's derivation. Lecture 9 writes A as samples × neurons; PN4 writes it as neurons × samples. The Gram matrix is neurons × neurons either way.

1. An accumulate-to-threshold decision model (PN4 §1)

Skill: Turn a recurrent network into its dynamics, then explain saturation, noise and the decision readout with sketches.

PN4 builds a 1-D, 50-neuron ensemble (radius 1) with a recurrent identity connection (synapse 0.1 s) and a constant input u connected with transform 1.

1a. Derive. Use the conventions to find A and B, ignoring the input synapse mismatch. Write $x(t)$ from $x(0) = 0$. The notebook asks "what is the mathematical computation here?"; answer it in one line.

1b. Calculate and graph. Find the time $t^*(u)$ at which the ideal x reaches the threshold 0.9. Evaluate it for $u = 0.1, 0.2, 0.3$. Sketch t^* against $|u|$, and connect the shape to the claim that "weaker evidence takes longer."

1c. Saturation, graphically. Decoders for the recurrent identity were fitted only on $[-1, 1]$. Sketch, on axes $x \in [-2, 2]$, the line $y = x$ and a plausible decoded feedback $\hat{f}(x)$. Show where $\hat{f}$ departs from $y = x$, and explain why it bends that way. Write $\tau\dot{x} = \hat{f}(x) - x + u$, and use your sketch to find the equilibrium for $u = 0.2$ graphically. Then sketch $x(t)$ for $u = \pm 0.1, \pm 0.2$ over 2 s.

1d. Noise. PN4 injects white noise into the ensemble with RMS $1, 3, 5, 10$. Write the resulting ideal model as "drift + noise." Sketch two noisy runs at $u = 0.1$ with small noise and two with large noise. Predict three effects of increasing noise on decision time and accuracy. Which psychological model of decision making does this become?

1e. Decision readout. Sketch choice(x) (threshold ± 0.9) and, on the same axes, a plausible version decoded from 50 broadly tuned LIF neurons. List three systematic differences between them. Why can a sum of smooth tuning curves not produce a jump?

1f. The intercept trick. The output ensemble now uses intercepts from $U(0.4, 0.9)$. Sketch tuning curves for four such neurons (two with $e = +1$, two with $e = -1$) over $y \in [-1, 1]$. Explain (i) why the output is now *exactly* zero until the decision, (ii) why the ± 1 values are represented better, and (iii) what is lost.

1g. One ensemble or two? PN4 replaces both 1-D ensembles by a single 2-D ensemble (100, then 500 neurons). Explain why the radius must grow, what happens to accuracy at 100 neurons, and why the intercept trick of 1f no longer works. What biological claim changes? Propose one experiment that could tell whether the brain keeps the two variables separate.

Before moving on: You can draw the decoded-feedback curve and point to the equilibrium it creates.

2. Representing a function as a vector (Lecture 7, first half)

Skill: Translate between a function, its samples and its basis coefficients.

Shifted Legendre polynomials on $r \in [0, 1]$: $\tilde{P}_0 = 1$, $\tilde{P}_1 = 2r - 1$, $\tilde{P}_2 = 6r^2 - 6r + 1$.

2a. Sampling. State the Nyquist–Shannon result in one sentence. How many samples completely determine a 2 Hz band-limited signal over a 0.5 s window? Over a 3 s window with a 5 Hz limit? Why is sampling a special case of a basis expansion?

2b. Orthogonality by hand. Show $\int_0^1 \tilde{P}_0 \tilde{P}_1 \, dr = 0$ and compute $\int_0^1 \tilde{P}_1^2 \, dr$. Is this basis orthonormal?

2c. Coefficients without integrals. Write $f(r) = r$ and then $f(r) = r^2$ exactly as combinations of $\tilde{P}_0, \tilde{P}_1, \tilde{P}_2$. Check your answers at $r = 0$ and $r = 1$. Which coefficients would a $q = 2$ representation of r^2 drop?

2d. Represent versus compute. In one or two sentences each, contrast *representing* a function in a population with *computing* $y = f(x)$ across a connection. Then explain why "solving for decoders" is the same least-squares problem as finding basis coefficients, and what plays the role of the basis.

Before moving on: You can explain why a q-dimensional vector can stand for a whole window of a signal.

3. The Legendre Memory Unit (PN4 §2, Lecture 7)

Skill: Derive small LMUs by hand, decode delays, and predict the effects of q and neuron count.

For $i, j = 0, \ldots, q-1$: $\theta A_{ij} = (2i+1) \cdot \{-1 \text{ if } i < j;\ (-1)^{i-j+1} \text{ if } i \geq j\}$ and $\theta B_i = (2i+1)(-1)^i$. The state approximates the window $u(t - \theta') \approx \sum_i \tilde{P}_i(\theta'/\theta)\, x_i(t)$ for $0 \leq \theta' \leq \theta$.

3a. $q = 1$. Compute A and B, and write the ODE. What familiar filter is it, and what does x hold about the past? Give A', B' for $\tau = 0.1$, $\theta = 0.5$. Sketch $x(t)$ for a unit step input at $t = 0$, and compare it with a true 0.5 s delay of the step.

3b. $q = 2$. Write the 2×2 matrix θA and the vector θB, then A' and B' for $\tau/\theta = 0.2$. What are the shapes of A' and B' for general q?

3c. Decoding delays. Using $\tilde{P}_i(0) = (-1)^i$ and $\tilde{P}_i(1) = 1$, write the decoding weights that read out the *current* input and the input θ *seconds ago* from a $q = 4$ state. Give the weights for the middle of

the window, $\theta' = \theta/2$, using your polynomials from Stage 2. Why must a system that implements a pure delay remember the entire window?

3d. The classification task. With $\theta = 0.5$ s, how many cycles of a 1 Hz and a 2 Hz sine fit in the window? Predict the window-average coefficient x_0 for each frequency. Explain why a 1-D input value cannot separate the classes but the 6-D state can.

3e. Order matters (PN4 2c). The classifier was fitted on a run with 1 Hz first. Run with 2 Hz first, it misclassifies for roughly the first 0.25 s. Explain two contributing reasons, involving the empty initial window and the training data, and propose two fixes.

3f. Sweeps (PN4 2d–e). Sketch the predicted classification RMSE against the neuron count ($50 \to 5000$, log axis) and against q ($1 \to 64$, log axis, 1000 neurons). Justify the shape of each in two sentences. Why do the LMU state components exceed 1 in magnitude, and what does that imply for the ensemble's radius or intercepts? Lecture 7 uses `CosineSimilarity(q+2)` intercepts; what problem does that solve?

Before moving on: You can derive the $q = 1$ LMU and explain why it is only a low-pass filter.

4. Online learning with PES (PN4 §3, Lecture 8)

Skill: Derive the rule, draw the network, and predict learning curves.

4a. Why learn? Lecture 8 defines learning as a directed change in synaptic weights while the network is active. Give two reasons to learn weights that NEF could otherwise solve for offline. Name one adaptive phenomenon that is *not* learning under this definition.

4b. Derive the delta rule. From $E = \frac{1}{2}(\sum_i d_i a_i - y^d)^2$, derive Δd_j. Interpret the sign: if neuron j is active while the output is too high, what happens to d_j?

4c. From decoders to weights. Using $w_{ij} = \alpha_i \langle \mathbf{e}_i, \mathbf{d}_j \rangle$, derive the vector PES rule. Identify which factor is presynaptic, which is postsynaptic, and which is the "error as input" (modulatory) term. Why is this more biologically plausible than the decoder rule?

4d. Draw the network. Draw PN4's identity-learning network: stimulus, pre, post and error ensembles, each connection's function and sign, and the connection into the learning rule. Modify it to learn $x \mapsto [x^2, -x]$: which ensembles change dimension, and where does the `function=` go?

4e. Sign error. Predict and explain the behaviour if the error is computed as `target - output`.

4f. Learning curves. Sketch the five 2-s chunk RMSEs for a 10 s run ($\eta = 10^{-4}$), then a 100 s run, then a 100 s run at $\eta = 10^{-5}$. Describe the speed versus final-noise trade-off and why the error never reaches zero. List three changes that would make the learned result more accurate.

4g. Forgetting. Suppose the input stays near $x \in [0.5, 1]$ for a long time. Using $\Delta d_j \propto \varepsilon a_j$, explain which decoders change and why the mapping for negative x can degrade (catastrophic forgetting).

4h. Where does the error come from? In Lecture 8's classical-conditioning model, which signal plays the role of the target, and how is the error generated internally? What happens during extinction?

Before moving on: You can write the PES rule from memory and explain every factor.

5. Adaptive control of a pendulum (PN4 §4)

Skill: Explain what an adaptive ensemble learns and why a controller output is a valid error.

Use $mL^2\ddot{\theta} = \tau - mgL\cos\theta$ and $\tau_{PD} = k_p(\theta^* - \theta) - k_d\dot{\theta}$, with θ measured from the horizontal.

5a. Offset. At equilibrium, derive an equation for the steady-state angle with PD alone. For a small offset, show that $\theta_{ss} \approx \theta^* - mgL\cos\theta^*/k_p$. Evaluate it for $\theta^* = 0$ and $k_p = 49$, and then for $\theta^* = \pi/2$. Why is the offset zero there? Why not just make k_p huge?

5b. What is learned. Write the torque the adaptive connection should supply as a function of state. Why must the adaptive ensemble represent θ, and not merely receive a constant input?

5c. The error. Show that, at any equilibrium, $\tau_{PD} = mgL\cos\theta - u_{ad}$. Use this to argue that driving $\tau_{PD} \to 0$ drives $u_{ad} \to mgL\cos\theta$. With $\varepsilon = \hat{y} - y^d$ and $\Delta d = -\eta\varepsilon a$, what sign must the PD output have in the error connection?

5d. Sketch. Sketch $\theta(t)$ from $\theta(0) = \pi/4$ for PD only and for PD plus adaptation. After learning, the target switches to $\theta^* = \pi/4$. Predict whether a large offset reappears, and explain why.

5e. Compare with Stage 4. In one sentence each, state what is the same as identity learning and what differs.

Before moving on: You can say what plays the role of the error without saying "the target."

6. Unsupervised learning (Lecture 8; not in PN4, so keep it brief)

Skill: Recognize Hebbian-family rules, their stability and their link to PCA.

6a. Write the basic Hebbian rule. Explain why it is unstable and why it ignores causation.

6b. Write Oja's rule and the BCM rule. For each, say in words how stability is achieved. To what does Oja's weight vector converge?

6c. Sketch the STDP window $W(\Delta t)$ against $\Delta t = t_{post} - t_{pre}$, labelling the potentiation and depression sides. (Check the exponent's sign for $\Delta t < 0$: the window must decay away from zero on both sides.)

6d. PCA finds the eigenvectors of which matrix? How does the SVD of the centred data give the same answer? Name one limitation of PCA relative to an autoencoder.

7. Analysing representation with the SVD (PN4 §5, Lecture 9)

Skill: Explain which functions a population supports, and why noise limits them, using equations and sketches.

7a. Which matrix? With N neurons and M samples, write the decoder equation and identify Γ and Υ with their shapes. Explain why decomposing Γ tells us about *every* function, and give two interpretations of what a large singular value means.

7b. SVD algebra. With $\Gamma = USU^T$ and $U^TU = I$, show that $\Gamma^{-1} = US^{-1}U^T$ and $(\Gamma + \sigma^2 I)^{-1} = U(S + \sigma^2 I)^{-1}U^T$. Explain in words why noise regularization "protects" the small singular values. What does a pseudo-inverse do with $S_m = 0$?

7c. Basis functions. Define $\chi = AU$ and $\Phi = U^T D$, and show $\hat{X} = \chi\Phi$. Sketch the spectrum (log axis) and $\chi_1, \ldots, \chi_4$ for 100 LIF neurons on $[-1, 1]$. Which classical polynomial family do they resemble? What would Gaussian (bump) tuning curves give instead? What happens to the ordering if all the "on" neurons have intercepts clumped near $+1$?

7d. Noise, derived. Starting from $E = \langle (x - \sum_m \chi_m \Phi_m)^2 \rangle_x + \sigma^2 \sum_m \Phi_m^2$ with $\langle \chi_m \chi_n \rangle_x = S_m \delta_{mn}$, set $\partial E / \partial \Phi_n = 0$ to obtain $\Phi_n = \langle \chi_n x \rangle / (S_n + \sigma^2)$. Explain what happens to the contribution of mode n when $S_n \gg \sigma^2$ and when $S_n \lesssim \sigma^2$. Then state the rule for which modes are usable (PN4 5c).

7e. Vectors. For a 2-D population, list the polynomial terms of orders 0–3 and count them. Compare your count with $N_S(l, D) = \binom{l+D-1}{l}$, and evaluate it for $D = 3$, $l = 2$. Why do higher-dimensional populations support high-order functions less well?

7f. Tie it together. Use the spectrum and the noise rule to explain three things you saw earlier: (i) why choice(x) in Stage 1 was decoded with rounded corners and under-reached ± 1; (ii) why PES learns x^2 more slowly than x (hint: under averaged gradient descent, each mode relaxes at a rate proportional to ηS_m); and (iii) why heterogeneous tuning is useful.

Before moving on: You can draw the spectrum with the noise floor and explain $S_m > \sigma^2$ in one sentence.

8. Independent readiness check (closed notes, about 20 minutes)

Skill: Retrieve the core reasoning without the scaffolding.

8a. A recurrent integrator uses $\tau = 0.05$ s, recurrent transform 1 and input transform 1. Find B, the ideal time to reach 0.9 for $u = 0.3$, and sketch the decoded trace, including saturation.

8b. Derive the $q = 1$ LMU and its A', B' for $\tau = 0.05$, $\theta = 1$. What does it remember about the past?

8c. Derive PES from the squared error. Draw the network that learns $f(x) = -x$, with signs.

8d. A PD-only pendulum with $k_p = 98$ holds $\theta^* = 0$. Find the offset and sketch PD versus adaptive control.

8e. Write the noise-optimal Φ_n, sketch the spectrum with the noise floor, and state which functions of x a 100-neuron LIF population can and cannot decode well.

Readiness rule. Mark each item *independent*, *needed a hint* or *not yet*. Redo the matching stage for any item that is not independent. Then take the 30-minute practice test under timed conditions.